2026. M81BC 2026L562A2EL

Coimisiún na Scrúduithe Stáit
State Examinations Commission

Leaving Certificate Examination, 2026

Design & Communication Graphics

Higher Level
Section B and Section C (180 marks)

Thursday, 18 June
Morning, 9:30 - 12:30

This examination has three sections:	
Section A	Core - Short Questions
Section B	Core - Long Questions
Section C	Applied Graphics - Long Questions

Section B	• Three questions are presented. • Answer **any two** questions on drawing paper. • All questions in Section B carry **60 marks** each.

Section C	• Five questions are presented. • Answer **one** question (i.e. the option you have studied) on drawing paper. • All questions in Section C carry **60 marks** each.

General Instructions:

- Construction lines must be shown on all solutions.
- The graphics presented are not necessarily drawn to scale and must not be used for scaling purposes.
- Write the question number distinctly on the answer paper in Section B and Section C.
- Work on one side of the drawing paper only.
- All dimensions are given in metres or millimetres.
- Write your Examination number in the box provided on Section A and on all other sheets used.

Do not hand this up.

This document will not be returned to the State Examinations Commission.

Section B - Core

Answer **any two** questions from this section on drawing paper.

B-1. The image on the right shows an ice bath.



Fig. B-1 shows an incomplete trimetric projection, using the axonometric axes method, of a similar ice bath. The elevation and plan are shown in their required positions.

A pictorial view of the ice bath is also shown.

(a) Draw the axonometric axes **X**, **Y**, and **Z** and the scalene triangle **abc**.

(b) Draw the elevation and plan, orientated as shown.

(c) Complete the trimetric projection of the ice bath.

(d) In the axonometric view, determine and indicate the horizontal and vertical traces of the simply inclined plane that contains surface **S**.

Scale 1:1

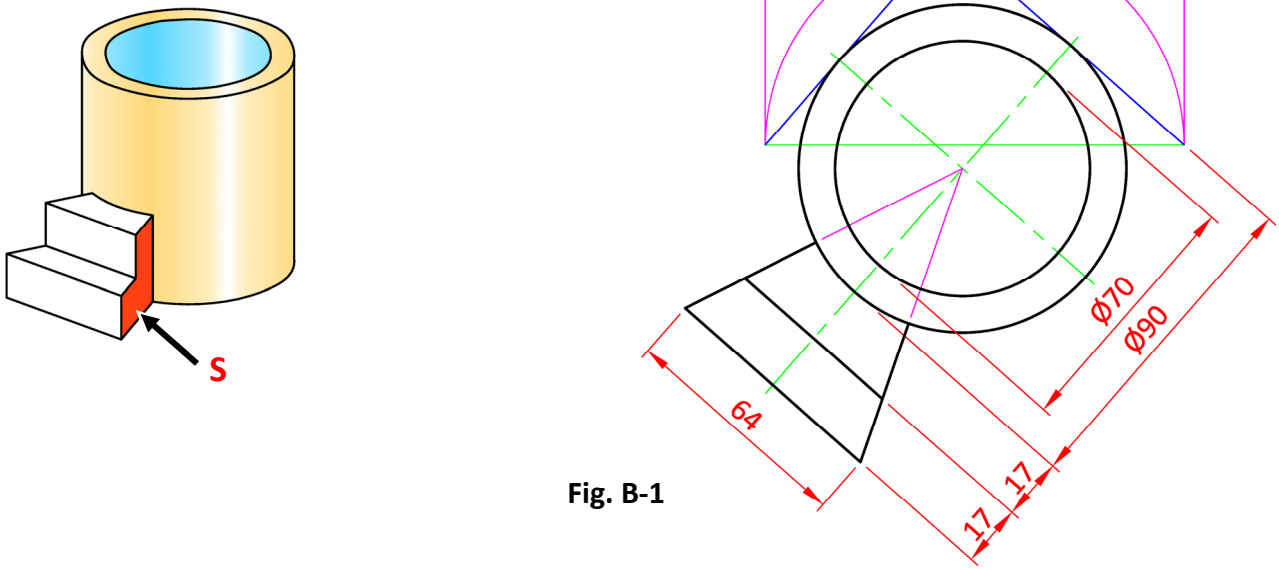


Fig. B-1

B-2. The image on the right shows a golf chipping net.

The chipping net consists of a circular base frame **A** and a circular target net **B** of the same diameter. The target net is inclined at 40° to the horizontal plane.

Fig. B-2 shows the plan, elevation, and auxiliary elevation of a similar chipping net. Also shown are the projections of an oblique plane **VTH** which contains the target net.

A pictorial view of the chipping net is also shown.

(a) Draw the base frame **A** and the horizontal trace in plan.

(b) Project the auxiliary elevation of the chipping net and complete the plan.

(c) In elevation:

(i) Determine the vertical trace.

(ii) Project the elevation of the chipping net.

(d) Determine the true angle between the **VT** and **HT**.



Scale 1:10

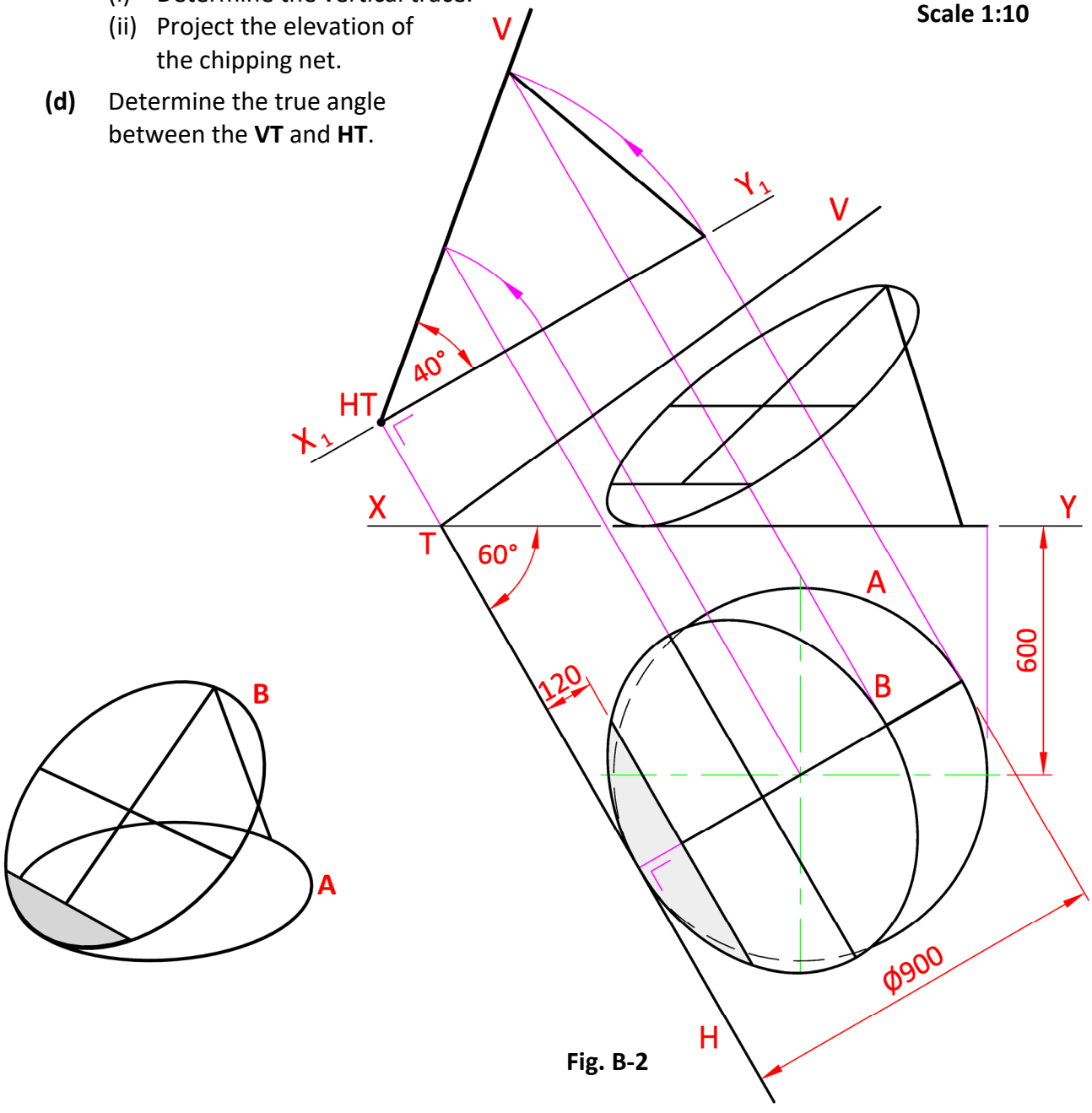


Fig. B-2

B-3. The image on the right shows a structure in Dublin Port. The geometry of the structure includes a series of planar triangular surfaces.

Fig. B-3 shows the plan and elevation of four intersecting planar surfaces.

The horizontal and vertical coordinates for points **A**, **B**, **C**, **D**, and **E** are given. Partial coordinates for the point **F** are also given.



Scale 1:1

A:	100	---	25	---	55
B:	150	---	20	---	70
C:	205	---	0	---	75
D:	230	---	60	---	20
E:	260	---	0	---	65
F:	280	---	?	---	30

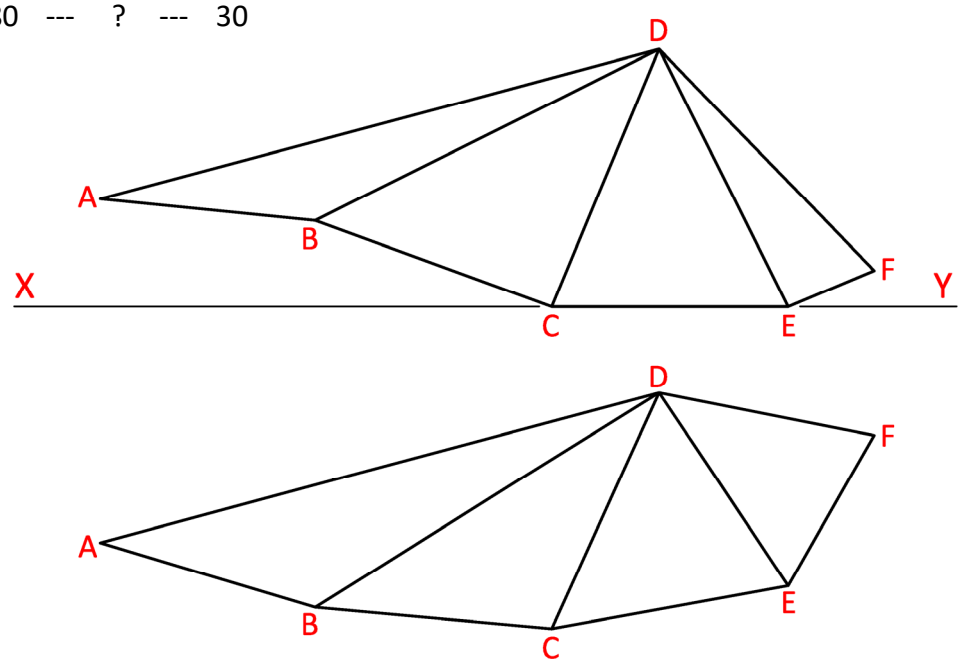


Fig. B-3

(a) Draw the given elevation and plan of the intersecting planes **ABD**, **BCD**, and **CDE**.

(b) Determine the dihedral angle between planes **ABD** and **BCD**.

(c) Determine and indicate the angle of inclination of the plane **CDE** to the horizontal plane.

(d) The dihedral angle between the planes **CDE** and **DEF** is 150°.
Complete the projections of the plane **DEF**.

Section C - Applied Graphics

Answer **one** question (i.e. the option you have studied) from this section on drawing paper.

Geologic Geometry

C-1. **(a)** The accompanying map, located on the back page of Section A, shows ground contours at five metre vertical intervals for a helipad and access roadway.



ABC is the centreline of the helipad and roadway. **A** is the centre of the circular curve which forms the helipad.

The helipad and roadway have the following specifications:

- the portion from **A** to **B** is level at an altitude of 40 m
- the portion from **B** to **C** is falling uniformly to a level of 35 m at **C**
- side slopes of 1 in 1.5 for the cuttings and 1 in 2 for the embankments.

Complete the earthworks necessary to accommodate the helipad and roadway on the northern side.

Note: The earthworks on the southern side of the helipad and roadway have already been completed.

(b) On a separate diagram on the map, the elevation and plan of three boreholes drawn from points **R**, **S**, and **T** are shown. They reveal the top surface of a stratum of ore at distances of 10 m, 12 m, and 22 m from **R**, **S**, and **T**, respectively.

(i) Draw the elevation and plan of the top surface of the stratum.

(ii) Determine the strike and dip of the stratum.

Scale 1:1000

C-2. The image on the right shows an office saddle stool.
The shape of the seat is based on a hyperbolic paraboloid.

The projections of a similar seat are shown in Fig. C-2 below.
The seat is formed by extending the hyperbolic paraboloid surface **ABCD**.

(a) Draw the given plan including the elements as shown.

(b) Project the elevation of the hyperbolic paraboloid surface **ABCD**.

(c) Complete the elevation by extending the elements of the hyperbolic paraboloid.

(d) Show the curvature of the surface when intersected by a vertical cutting plane passing through **B** and **D**.



Scale 1:2

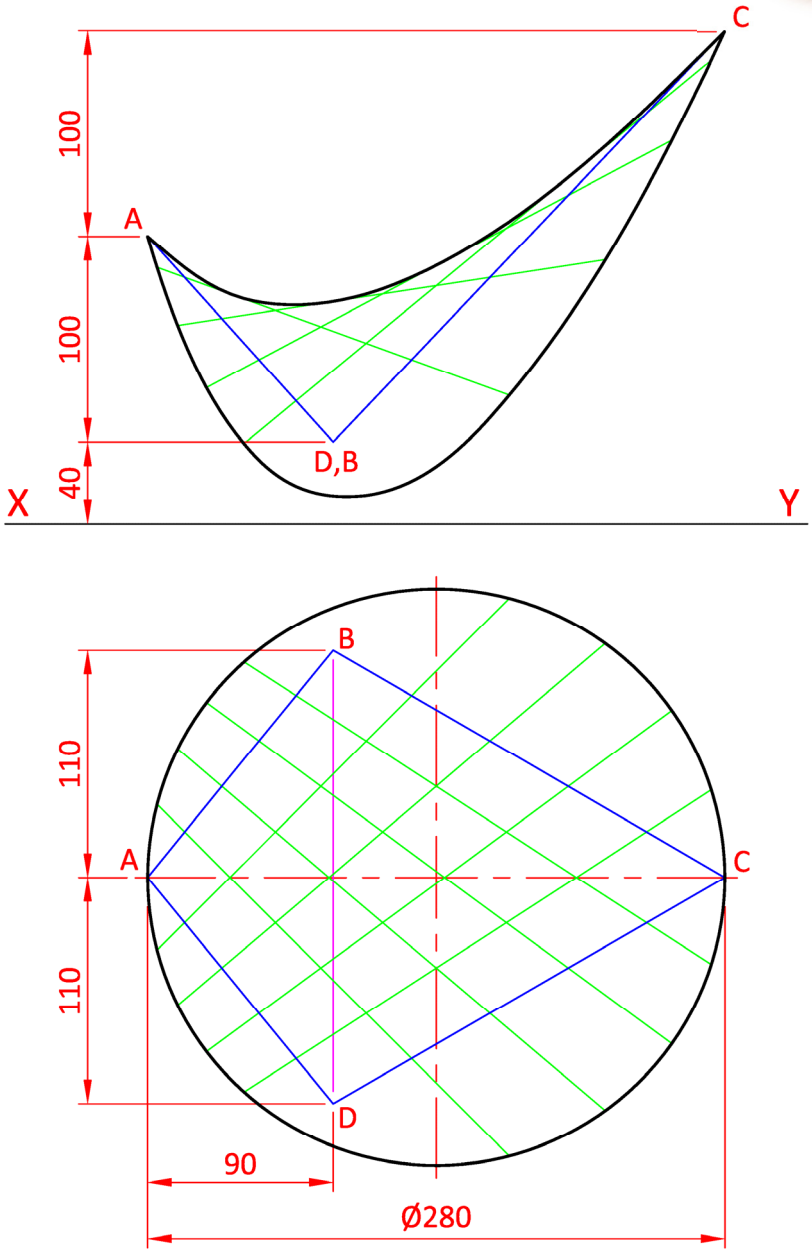


Fig. C-2

C-3. The image on the right shows the Vitra cone chair, by Verner Panton. The chair consists of a truncated cone.



Fig. C-3 shows the elevation and plan of a similar chair.

(a) Draw the given elevation of the chair.

(b) Project the plan.

(c) Draw a one-piece surface development of the conical surface **A**.

Scale 1:10

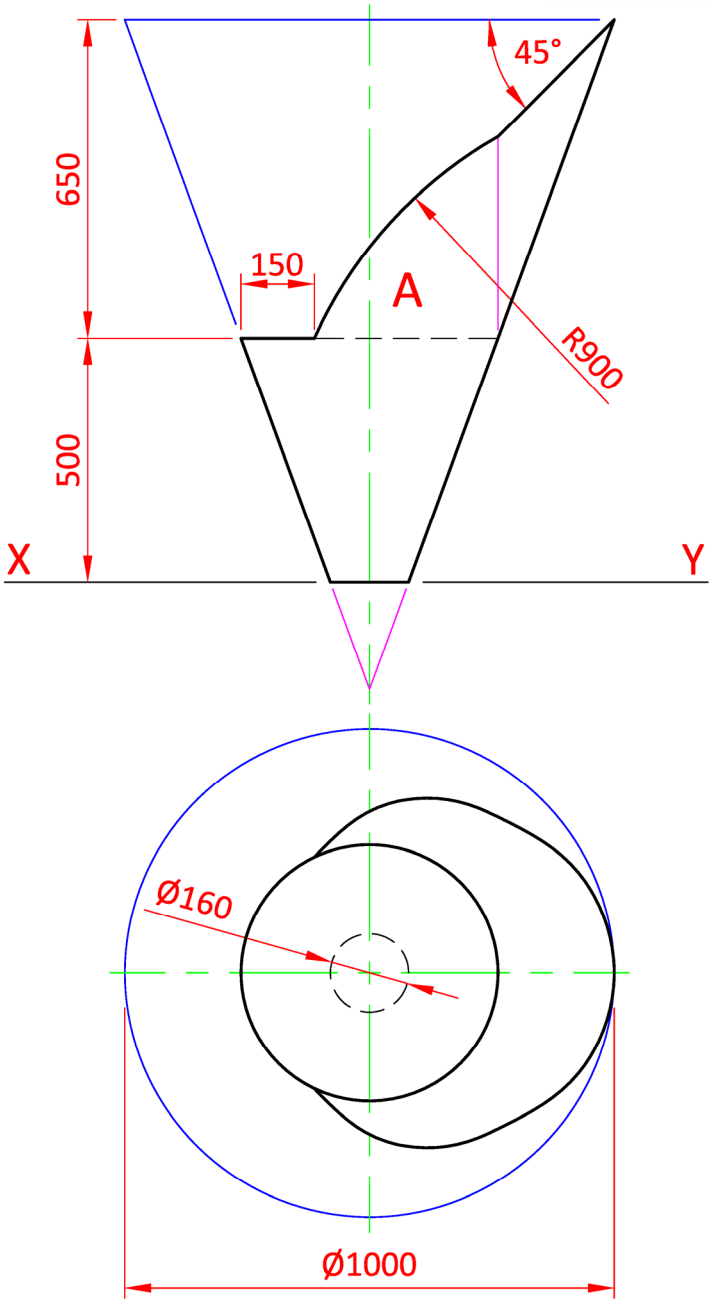


Fig. C-3

C-4. **(a)** The image on the right shows a peristaltic pump.



Scale 1:1

A peristaltic pump uses a cam to compress a flexible tube which forces a liquid inside to move in one direction.

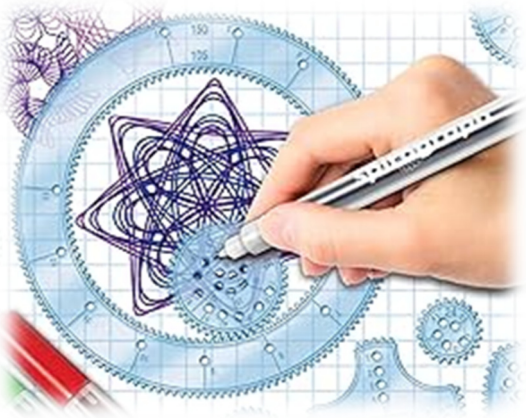
Draw the displacement diagram and profile of the radial plate cam which imparts the following motion.

- 0° to 180° rise of 60 mm with simple harmonic motion
- 180° to 300° dwell
- 300° to 360° fall 60 mm with uniform velocity.

The nearest approach of the tube to the cam centre is 30 mm.
The camshaft diameter is 24 mm. The cam rotates in a clockwise direction.

Use a distance of 10 mm for each 30° interval on your displacement diagram.

(b) The image on the right shows a spirograph game. The game creates geometric patterns.



Scale 1:1

In the spirograph, a pen is placed in a hole in the small circular wheel. Rolling the wheel along the inside of the larger circular ring traces a curve.

Fig. C-4 below shows a circle **C** and an arc **AB** representing the wheel and the ring.
P is a point located within the circular wheel.

(i) Draw the given diagram.

(ii) Plot the locus of point **P** as the circle **C** rolls along the arc **AB**, without slipping, for one complete revolution.

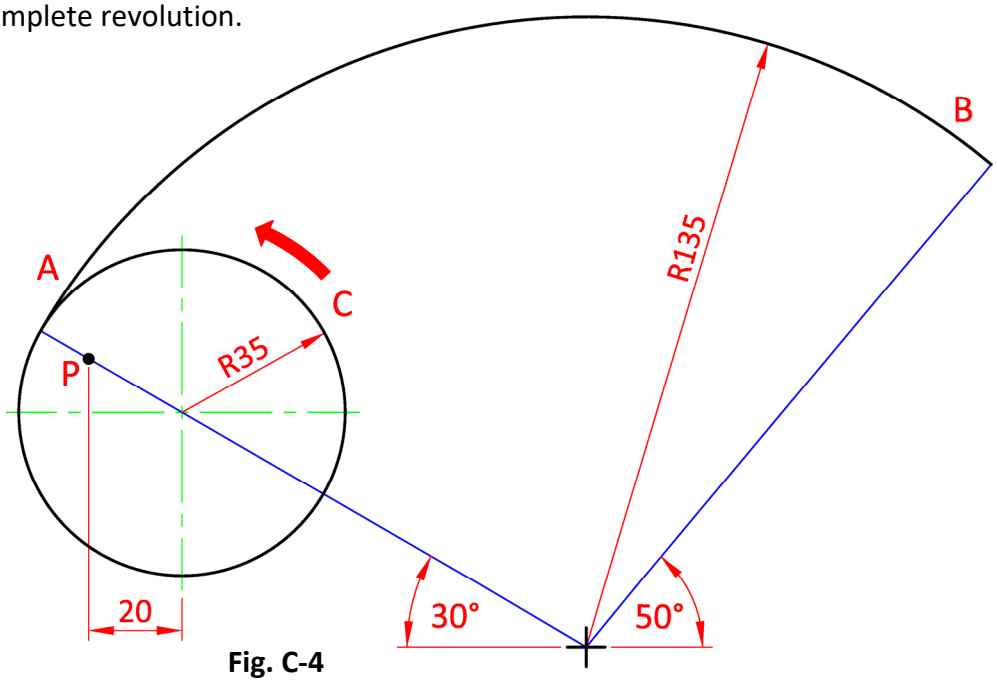
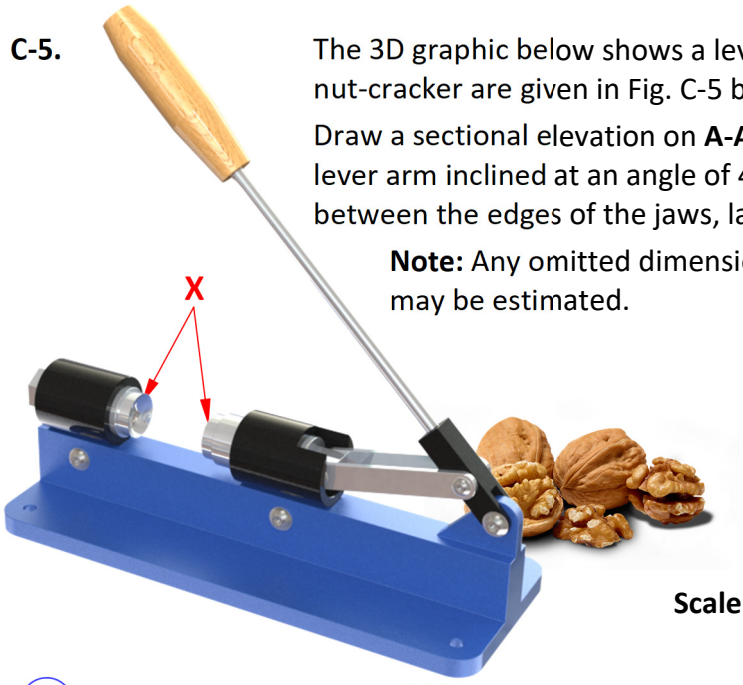


Fig. C-4

Assemblies

C-5.



The 3D graphic below shows a lever nut-cracker. The details of the nut-cracker are given in Fig. C-5 below. The parts list is also given in a table.

Draw a sectional elevation on **A-A**, showing the parts fully assembled with the lever arm inclined at an angle of 45° to the horizontal plane and the distance between the edges of the jaws, labelled **X**, equal to 26 mm.

Note: Any omitted dimensions may be estimated.

Scale 1:1

Part	Name	Qty.
1	Main Body	1
2	Adjustable Jaw Guide	1
3	Sliding Jaw Guide	1
4	Link Arm	2
5	Lever Arm	1
6	Adjustable Jaw	1
7	Sliding Jaw	1
8	M3 × 14 Set Screw	2
9	M3 Plain Nut	2
10	M4 × 14 Set Screw	3
11	M4 Plain Nut	1

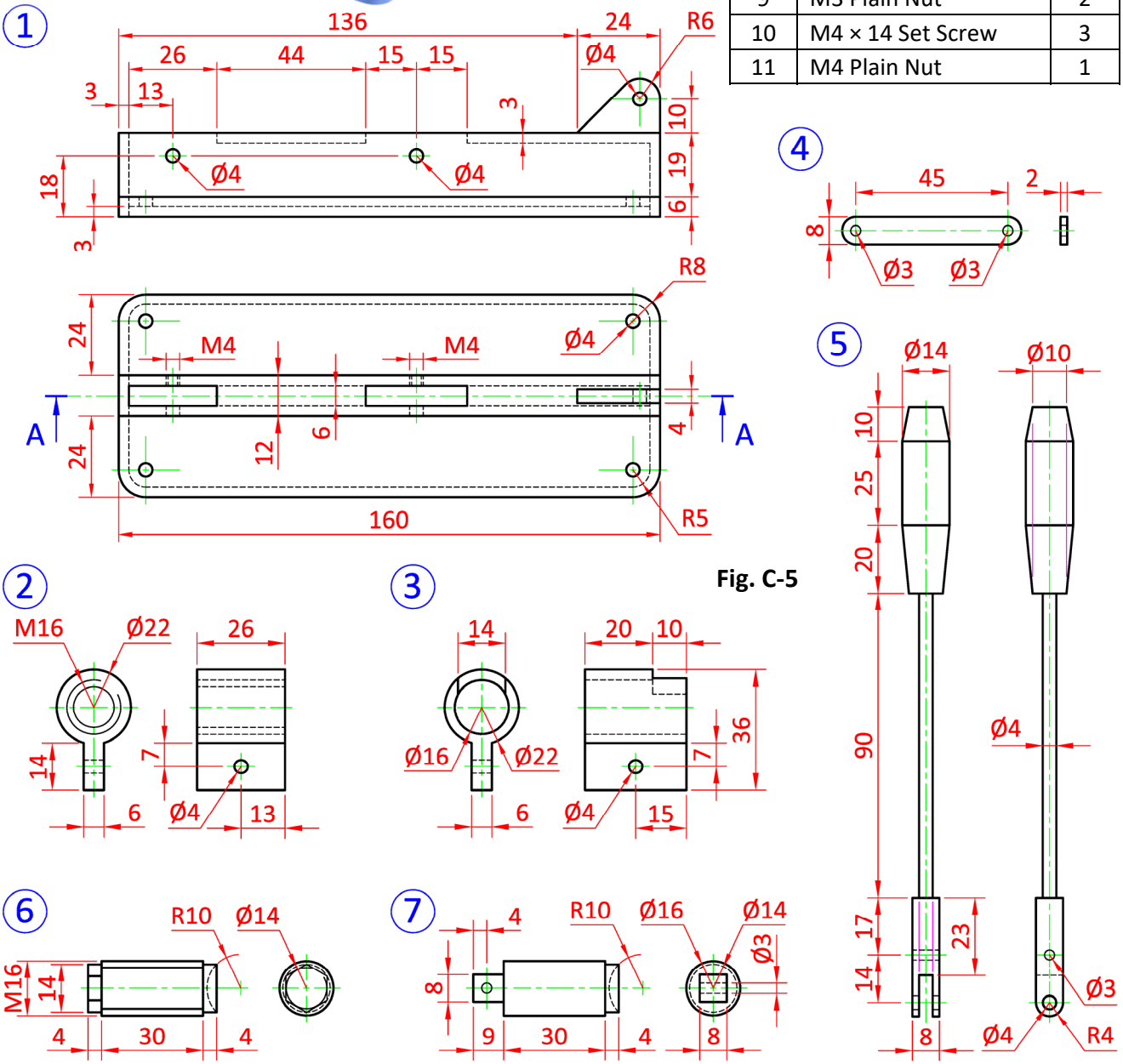


Fig. C-5

There is no examination material on this page

There is no examination material on this page

Do not hand this up.

This document will not be returned to the State Examinations Commission.

Leaving Certificate - Higher Level

Design & Communication Graphics, Section B & Section C

Thursday, 18 June
Morning 9:30 - 12:30